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CMSE 402

Final Write-up

Two datasets were used for this study in order to analyze the connection between income inequality and obesity in US counties. The first one is the USDA Food Environment Atlas which contains information on income, food environment, and obesity rate of each county. The other dataset that will be used in the analysis is the County Health Rankings dataset produced by the Robert Wood Johnson Foundation. It includes data from 2010 to 2025 for all the counties of the country.

Four primary questions shaped my analysis: Is there a correlation between poverty and obesity? Is the same pattern applicable to other health indicators? Did the wealth disparity between rich and poor counties change over time? Where does this issue persist in the US? The answer to each query is unequivocally affirmative. My conclusion is that obesity in the United States is not the fault of the individuals but rather the result of a structural problem that is becoming increasingly severe.

From a technical perspective, the most valuable takeaway for me was just how much effort is required when preprocessing raw data. The County Health Rankings dataset contained 16 years of data that needed to be cleaned because the file structures did not match the data, the formats were inconsistent, and the FIPS codes differed among the data entries. Designing a data cleaning process to tackle these problems proved to be more complicated than anticipated, and it taught me a lesson that will stick with me for future endeavors – always familiarize yourself with your data prior to visualization.

With regards to the design aspect, one of the most useful things I found out during my project was the effort required to create a readable chart at poster size. Fonts, colors, and margins that looked perfectly good on my computer monitor failed at scaling up to a big poster. Once I finally had a decent looking presentation I switched to Canva which I found as a great tool.

Another lesson I took away was that argument is as important as analysis. This point came out quite strongly when reviewing my draft because beautiful figures without a strong argument just look like data dumps, not a project. Organizing the entire project according to a single argument made a lot of difference and forced me to rethink the purpose of each figure.

It was no surprise to see a negative correlation between income and obesity; in other words, the income-obesity relationship was as expected. For all years included in the data set, the correlation remained negative. It was also no surprise to see regional clusters in the southern states and Appalachian region.

The surprising aspect for me was the rise in the bottom quintile counties after 2021. It was anticipated that there would always be a difference between the quintiles, but the extent to which the difference grew in a very short span of time was not expected. This was one of the strongest arguments that emerged from the study, and initially, I had not thought about it.

Data cleaning was not an easy task either. At first glance, combining data from multiple years seemed to be a relatively easy job, but that was not the case. The column name differed among the years, while FIPS code existed in different forms. Moreover, some attributes were represented in terms of proportions in some years while percentages in others.

The last thing I didn't expect was the tedious work of getting a poster to look good. Making the images big enough to be legible while making sure that everything is included is a true design problem, one that has led me to learning about designing the poster via code.